



S. No. ** 0098416 **

AIR UNIVERSITY, ISLAMABAD

ANSWER BOOK: MID SEMESTER EXAMINATION

REGISTRATION ID. 15922 CLASS & SEC. 0098416 SUBJECT ARTIFICIAL INTELLIGENCE
S.NO. OF EXTRA SHEET/S (IF USED) 1. _____ 2. _____ 3. _____ 4. _____ 5. _____ 6. _____ 7. _____
EXAMINATION (Please ✓ the appropriate Box) ☒ FALL ☐ SPRING ☐ SUMMER 20 25
SIGNATURE OF STUDENT [Signature] DATE 08-Dec-2025
SIGNATURE OF INVIGILATOR [Signature] NAME OF INVIGILATOR Malik Adnan

IMPORTANT INSTRUCTIONS

BEFORE STARTING YOUR WORK, ENSURE THAT:

1. YOU ARE IN POSSESSION OF YOUR PRINTED ROLL NUMBER SLIP AND AU ID CARD.
2. YOU ARE SITTING IN THE ROOM AND SEAT ASSIGNED TO YOU AS PER THE ROLL NUMBER SLIP.
3. YOU DO NOT HAVE ANY UNAUTHORISED PRINTED OR WRITTEN MATERIAL, CELL PHONES, SMART DEVICE OR ANY OTHER MATERIAL THAT MAY BE AN UNFAIR AID AS PER AU ACADEMIC REGULATIONS. DISCOVERY OF ANY SUCH MATERIAL AFTER START OF THE EXAMINATION WILL RENDER YOU LIABLE FOR DISCIPLINARY ACTIONS UNDER CHAPTER XV OF AU REGULATIONS.
4. WRITE YOUR REGISTRATION ID AND PARTICULARS CAREFULLY ONLY IN THE SPACES SPECIFIED ABOVE FOR THIS PURPOSE.

DURING THE EXAMINATION:

5. ENSURE THAT ANSWER BOOK AND ALL CONTINUATION SHEETS ARE SIGNED BY THE INVIGILATOR.
6. ANSWERBOOK CONTAINS 10 PAGES. TEARING OF ANY PAGE FROM THE ANSWERBOOK OR CONTINUATION SHEET IS AN OFFENCE UNDER AU UNFAIR MEANS POLICY.
7. DO NOT WRITE ANYTHING ON QUESTION PAPER EXCEPT YOUR REG ID. USE LAST PAGES OF ANSWERBOOK FOR ROUGH WORK.
8. DO NOT BORROW ANYTHING FROM OTHERS. RAISE YOUR HAND TO SEEK ASSISTANCE OF THE INVIGILATOR.
9. DO NOT USE LEAD PENCIL (EXCEPT FOR DIAGRAMS/GRAPHS ETC.) OR ANY INK OTHER THAN BLUE OR BLACK.
10. WRITE ON BOTH SIDES OF EACH PAGE IN THE ANSWERBOOK.
11. ENSURE THAT SERIAL NUMBERS OF ANSWERBOOK AND CONTINUATION SHEETS ARE ENTERED ON ATTENDANCE SHEET. ALSO ENTER SERIAL NUMBER OF ALL CONTINUATION SHEETS IN THE SPACE PROVIDED ON THE ANSWERBOOK.

AFTER THE EXAMINATION:

12. MAKE SURE THAT ALL CONTINUATION SHEETS ARE STAPLED WITH MAIN ANSWERBOOK AT THREE PLACES BEFORE SUBMITTING YOUR ANSWER SCRIPT TO THE INVIGILATOR.
13. DO NOT LEAVE EXAM HALL WITHOUT PERMISSION OF INVIGILATOR EVEN IF YOU HAVE COMPLETED THE EXAM.
14. IN NO CASE SHOULD YOU LEAVE THE EXAM HALL BEFORE HALF TIME OR YOUR PAPER MAY BE CANCELLED.

Q. NO.	1	2	3	4	5	6	7	8	9	10	TOTAL
MAXIMUM MARKS	20	30									50
MARKS OBTAINED	12.5	9.10									22.5

MARKS OBTAINED IN WORDS Twenty Two point five

EXAMINER'S SIGNATURE [Signature]

QUESTION 1:

(5)

a) ARTIFICIAL INTELLIGENCE:-

Artificial Intelligence making a decision, goal setting of the environment and help of the humans and decision making skills and provide easy and simple solution of each and every things.

Example:-

- 1) Automatic cars
- 2) Vacuum cleaner to check the dust and on the automatic
- 3) Rain sensor to predict the rain and observed and change the every moments

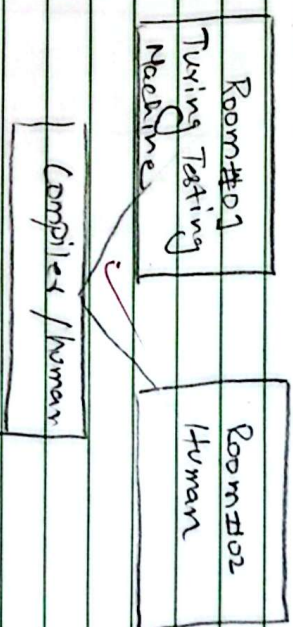
Turing Test:-

- 1) Purpose of Turing testing machine to help of the maximum observations and sets of different places and different observations.

- 2) The Turing testing machine to test the equality of checking and three empty rooms one room is AI machine/Turing testing machine and one of the humans and the human calls the question and the two

Simple answers of the two rooms one of Turing machine and one of humans.

Diagram:-



b) AGENTS:-

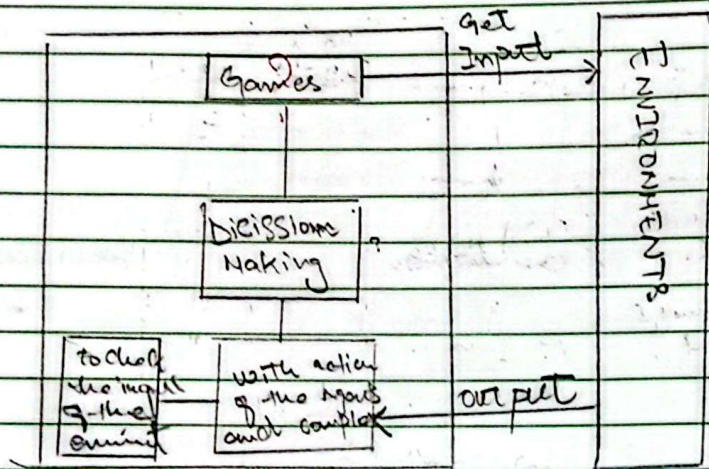
(2.5)

An agent the main purpose of Agents to the middle term basically the agents to the work of to given statements and check the sense work and check the work is properly and check the all instructions

Example:-

- 1) Vacuum cleaner check the dust and sense and start the vacuum its a task of the Agents of ensure each and every things-

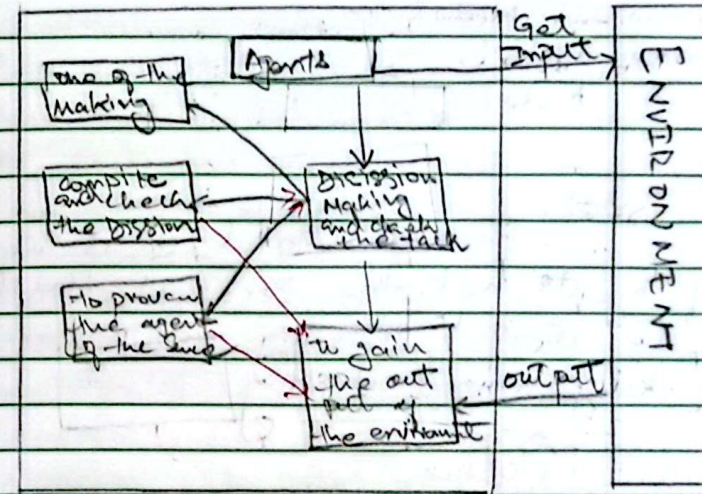
c) SIMPLE REFLEX AGENT: (2.5)



Define:

Simple reflex agents is the types of agent and its uses is very simple and easy to work and change the context of the reflections to get the input and the answers of the output to check the environment and get input to output of the simple reflex.

GOAL-BASED AGENT:



Define:

The goal base agents to check the environments skills and check the decisions of the task as per environments and the Agents takes action to check the three goals sets one by one and the prefer the best choice of the decision and get the input and the answers of output its goal base Agents.

d) PEAS MODEL:

2.5

P $\Rightarrow$ Personal ModelE $\Rightarrow$ Environment Model.A $\Rightarrow$ Actuator Model.S $\Rightarrow$ Sensor Model.

PERSONAL MODEL:

Personal Model it is a type of agents and the Personal Model to check the oriented checking and personal used of daily humans tasking.

ENVIRONMENT MODEL:

Environment's model is the type of Agents and its help to check the environment's actuators and sense the e.g. weather and Rainfall, flood checking and sense easily.

ACTUATOR MODEL:

Actuator's Model is the type of agents and its help of the main purpose of machine and tasking of daily performance and help out the machine tasking.

SENSOR MODEL:

Sensors Model to sense the environment and sense the lights for example L.E.D light is on every early morning to sense the sense and morning sunlight is of and the every night to sense the night and automatic on its a sensors model's work.

A. ARTIFICIAL INTELLIGENCE:

Artificial Intelligence making a decision, goal setting of the environment and help of the humans problems solving and paths of ways of humans and its a daily multitasking easily performed.

TERMINOLOGIES:

Terminologies of (AI) is the type of Artificial Intelligence its help of the terms, activities and chaotic tasking easily and with all forms of (AI).

CHARACTERISTICS AND ITS APPLICATION:

- 1) To check the sensors for example vacuum cleaners.
- 2) Its help to daily multitasking.
- 3) Its help to provided sufficient information.
- 4) Its provide to various information.

Q2 Explain

(8)

UNINFORMED SEARCH:

Uninformed search is the type of data structure and its used of the Artificial Intelligence. Uninformed search is a locally used to manage the data in trees shapes and define the shortest path of the goal state and its help to usually square houses and for example.

Example:

- 1) Google maps is the current location and the drop location to find the uninformed search and help to check location and easiest location find.

DEPTH-FIRST SEARCH (DFS):

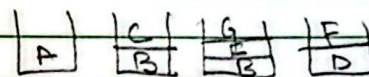
DFS to check the values of start state to end state the check the deepest value of the algorithms and its help the faster than BFS and check the data in few seconds and finds the shortest path of the goal state and work in easily.

Examples:

It is also used to the mainly searching and maps and k-maps searching to apply to this algorithms.

Explain1) COMPLETENESS:

The completeness of DFS is step by step?

1) TIME COMPLEXITY:

Its time complexity

$$O = (O+V)$$

$$O = (b^d)$$

$$O = (2)^2$$

$$O = 4$$

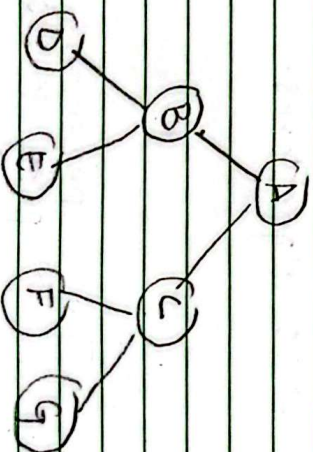
Ans1) Optimality:

The final path of Optimality

A → C → D → E → F → G

Data Structure = ?

Apk's Dfs



1. $\frac{1}{2}$
 2. $\frac{1}{3}$
 3. $\frac{1}{4}$
 4. $\frac{1}{5}$

$$4 = \sqrt{9} = 3$$

Linal Path :-

$$A \rightarrow B \rightarrow C \rightarrow D \rightarrow E \rightarrow F \rightarrow G$$

Ans



S.No

0064764

REGISTRATION ID 15922 SUBJECT AL S.NO. OF ANS. BOOK 5098416

MID SEMESTER EXAMINATION (Please ✓ the appropriate Box) ☒ FALL ☐ SPRING ☐ SUMMER 20 25

STUDENT'S SIGNATURE _____ INVIGILATOR'S SIGNATURE _____ DATE 18-Dec-25

5) Explain

Uniform-Cost Search (UCS)

22

Uniformed
cost search uses the working is the tree
is the shortest cost and the goal state
cost search of algorithms and its help
to the shortest cost works easily.

lets to manage and open the algo of the define working easily and help to work

2

1) COMPLETENESS

$S \rightarrow B \rightarrow M$

2) OPTIMALITY

$S \rightarrow A(1) B(4) C(3) D(5) E(5) F(4) G(2) H(1)$

$S \rightarrow A \rightarrow B \rightarrow G$

$S \rightarrow B \rightarrow M$

3) Time Complexity

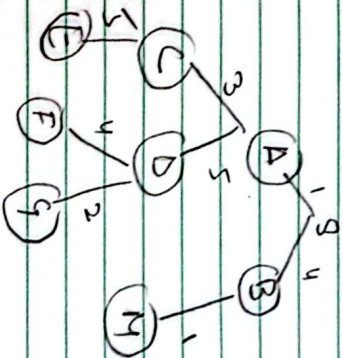
$$O = (O + V)$$

$$O = (E^4)$$

$$O = (2)^2$$

$$O = 8$$

3



$A(1), B(4), C(3), D(5), E(5), F(4), G(2), H(1)$

$S \rightarrow A \rightarrow B \rightarrow G$

$S \rightarrow B \rightarrow M$

$S \rightarrow B \rightarrow M$